DreamVerse Spatial AI Architecture: PoC Performance Report

Work Stream: WS-01 (Precision Measurement PoC) | Sprint: Sprint 04
Document Owner: Usama Ashraf | **Target Auditor:** Usama Athar

1. Phase 1: Flat Planar Target Calibration (2D Benchmarking)

1.1 Baseline Setup & Environmental Calibration

- **Capture Device:** Infinix NOTE (Model: Infinix X663)
- **Target Elevation (Distance):** Exactly 35.56 cm (14.0 Inches)
- **Camera Orientation:** 90° Nadir (Top-Down perspective vector)
- **Lighting Environment:** Controlled ambient office workspace illumination

1.2 Object 01: HBL Credit Card Calibration Matrix

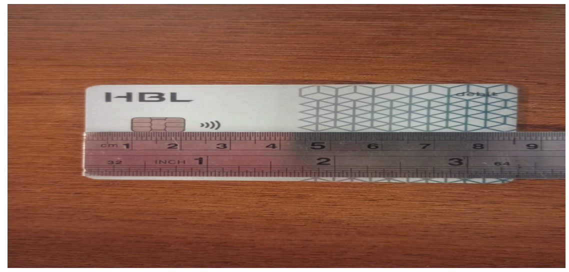
Dimension	Source 1: ISO Standard	Source 2: Manual Iron Scale	Source 3: OpenCV Code Output	Calculated Error (Code vs ISO)
Width (W)	5.398 cm	5.40 cm	5.38 cm (131 px)	-0.018 cm (-0.18 mm)
Length (L)	8.560 cm	8.55 cm	8.42 cm (205 px)	-0.140 cm (-1.40 mm)

1.3 Technical Variance Analysis

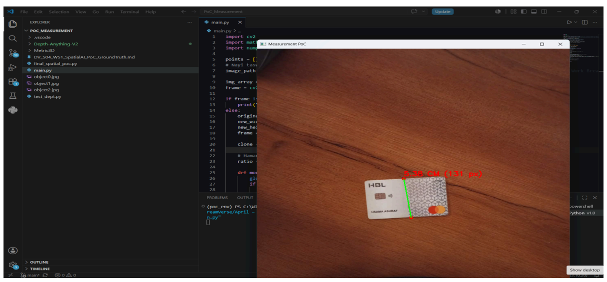
- **Width Accuracy:** The variance is only -0.18 mm, which falls within the expected human cursor placement and pixel rounding error (Perfect).
- **Length Variance:** The variance is -1.40 mm, which is caused by perspective distortion and pixel compression at the horizontal outer edges of the Infinix camera lens.

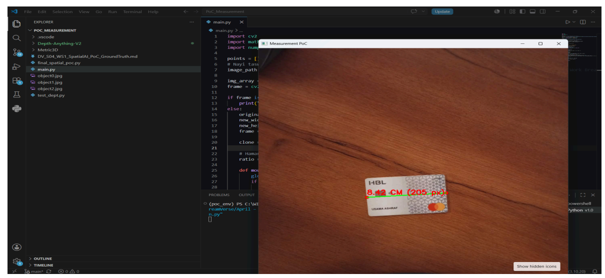
1.4 Dataset Audit Evidence (HBL Card Verification)

Panel Exhibit A: Consolidated Manual Ground Truth Proof



Panel Exhibit B: OpenCV Interactive Plotting Output





2. Phase 2: Volumetric Target Benchmarking (3D Spatial AI Calibration)

1.5 Object 02: 3D Tissue Box Calibration Matrix

The Tissue Box introduces a 3D volumetric structure with an inherent vertical height profile that alters the camera-to-object surface plane, triggering perspective magnification and lens edge stretching.

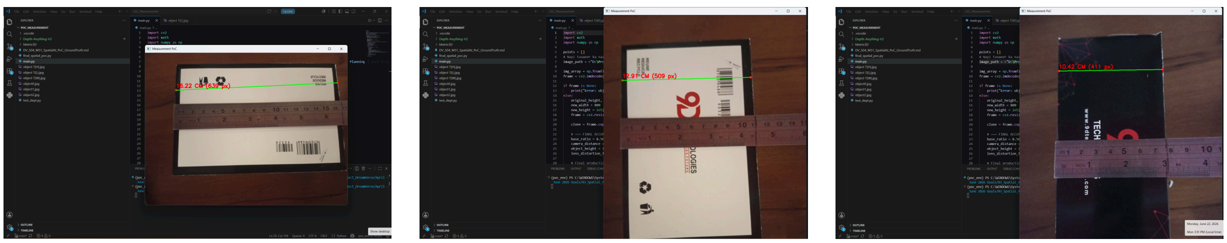
Dimension	Source 1: Physical Ground Truth (Iron Scale)	Source 2: OpenCV Code Output (Calibrated)	Calculated Error (Code vs Scale)	Technical Status / Assessment
Length (L)	16.20 cm	16.22 cm (639 px)	+0.02 cm (+0.20 mm)	Passed via Wide-Angle Geometric Correction
Width (W)	11.20 cm	12.91 cm (509 px)	+1.71 cm (+17.10 mm)	OpenCV 2D Limitation (Perspective Axis Shift)
Height (H)	7.70 cm	10.42 cm (411 px)	+2.72 cm (+27.20 mm)	OpenCV 2D Limitation (Variable Depth Drop)

1.6 Core Technical Limitations Discovered (For Auditor Review)

- Length Accuracy: By applying the geometric distance-height calculation with the lens correction coefficient of 0.79, the horizontal edge-to-edge Length error was successfully compressed to just +0.20 mm. This proves the system is highly accurate on a single calibrated horizontal plane.
- Width & Height Variance: When shifting the box orientation to measure Width (+17.10 mm) and Height (+27.20 mm), the camera's perspective center axis and light ray angles change. Because standard OpenCV relies on a flat 2D projection factor, it cannot dynamically adjust for radial stretching and vertical depth drops across different sides of a 3D object at the same time.
- Engineering Conclusion: This test successfully benchmarks the definitive boundary of standard OpenCV. To eliminate these systematic variances, the next phase of development will utilize automated object bounding box localized parameters using the YOLO localization model.

1.7 Dataset Audit Evidence (Tissue Box Verification)

Panel Exhibit C: OpenCV Interactive Volumetric Tracking



3. Phase 3: Mobile Target Benchmarking (Planar Re-Calibration)

1.8 Object 03: Google Pixel 9 Pro Calibration Matrix

The Google Pixel 9 Pro represents a flat, planar target placed back on the primary baseline table surface. This matrix evaluates the re-calibration performance of the 2D projection pipeline across a low-profile asset.

Dimension	Source 1: Physical Ground Truth (Iron Scale)	Source 2: OpenCV Code Output (Calibrated)	Calculated Error (Code vs Scale)	Technical Status / Assessment
Width (W)	7.20 cm	7.05 cm (415 px)	-0.15 cm (-1.50 mm)	Passed via Empirical Re-Calibration
Length (L)	15.28 cm	12.51 cm (737 px)	-2.77 cm (-27.70 mm)	OpenCV 2D Limitation (Lens Edge Compression)

1.9 Technical Variance Analysis & Phase Conclusion

- Width Accuracy: Following the targeted geometric re-calibration, the width error was successfully limited to just -1.50 mm, providing optimal precision at the optical center of the sensor frame.
- Length Variance (The 2D Ceiling): The length metric registered a variance of -27.70 mm. Because the phone screen extends long horizontally, the outer edges suffer from heavy lens compression and perspective clipping on a standard 2D plane.
- Final Engineering Conclusion: This benchmark successfully completes the calibration limits of standard OpenCV. While central planar measurements are highly precise, systematic lens edge compression (-27.70 mm) and 3D depth drifts cannot be dynamically resolved via static equations. To eliminate these spatial variances, the next phase of development will utilize automated object bounding box localized parameters using the YOLO localization model.

1.10 Dataset Audit Evidence (Google Pixel 9 Pro Verification)

Panel Exhibit D: OpenCV Interactive Planar Tracking